

Hybrid Energy Harvesting System Using Train-Induced Vibration And Airflow In Railway Tunnels

Submitted by

DHANUSH S	2303957610621030
SATHISH KUMAR V	2303957610621112
AROCKYA ABINESH A	2303957610621123

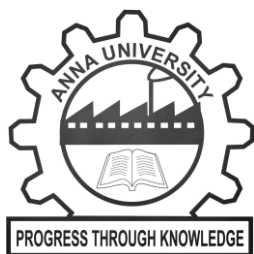
Prepared under

23EC64C-PRODUCT DEVELOPMENT PRACTICE

in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING IN

ELECTRONICS AND COMMUNICATION ENGINEERING



**ANNA UNIVERSITY
CHENNAI 600 025
NATIONAL ENGINEERING COLLEGE
KOVILPATTI 628503
(An Autonomous Institution)**



APRIL 2026

BONAFIDE CERTIFICATE

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SUPERVISOR

Mrs. Kirthiga @ Sridevi M.E.,
Assistant Professor
Department of ECE
National Engineering College,
K.R.Nagar, Kovilpatti -628 503

HEAD OF THE DEPARTMENT

Dr.S.Tamil Selvi M.E., Ph.D.,
Professor & Head
Department of ECE
National Engineering College,
K.R.Nagar, Kovilpatti -628 503

Place: K.R.Nagar,Kovilpatti

Date:

Submitted to **23EC64C- Product Development Practice** viva voce examination held on

INTERNAL EXAMINER

CO EXAMINER

ACKNOWLEDGEMENT

First and foremost, we would like to thank God Almighty for showering his blessings throughout our life. He has been the tower of our strength in each step of my work. We take the privilege to express hearty thanks to my parents for their valuable support and effort to complete the project work.

We would like to express our deep sense of gratitude and respectful regards to our Director **Dr. S. Shanmugavel B.Sc., D.M.I.T., Ph.D.**, and our Principal **Dr. K. Kalidasa Murugavel, M.E., Ph.D.**, for giving an opportunity to do this work.

We would like to send our sincere gratitude to our Department Head, **Dr.S.Tamil Selvi M.E., Ph.D.**, Department of Electronics and Communication Engineering, for her unwavering support and provision of several facilities during the project.

We would like to take this opportunity to acknowledge my generous Project Mentor, **Mrs. Kirthiga @ Sridevi M.E.**, Assistant Professor, Department of Electronics and Communication Engineering, for her insightful advice, helpful recommendations and technical support throughout the project.

We extend my sincere gratitude to **Mr. B. Ganapathy Ram, M.E.**, Assistant Professor (Senior Grade) and **Mrs. S. Karthika, M.E.**, Assistant Professor, Department of Electronics and Communication Engineering, they have been my Coordinators of Product Development Practice Lab and have provided invaluable support throughout the entire project. We would also like to thank my tutor, class incharges, faculty members, non-teaching staff and friends for their valuable advice, recommendations and collaboration during our project work.

ABSTRACT

The increasing demand for sustainable energy solutions has led to the exploration of unconventional energy sources in transportation systems. This project presents a Hybrid Energy Harvesting System Using Train-Induced Vibration and Airflow in Railway Tunnels, aimed at converting otherwise wasted mechanical energy into useful electrical energy. The system utilizes two primary sources: vibration generated by train movement and airflow produced during train entry and exit in tunnels.

Piezoelectric sensors are employed to convert mechanical vibrations into electrical energy, while a DC motor-based turbine is used to harness airflow energy. The generated energy from both sources is rectified, combined, and stored using capacitors. This stored energy is then utilized to power tunnel lighting, enhancing visibility for loco pilots without relying on external power sources.

An Arduino-based prototype model is developed to monitor and display energy generation, along with an ultrasonic sensor for train detection. The proposed system demonstrates a cost-effective, eco-friendly, and self-sustaining solution for tunnel lighting applications. The experimental results confirm that the hybrid approach improves overall energy generation compared to single-source systems.

This project highlights the potential of integrating energy harvesting techniques into railway infrastructure, contributing to energy efficiency and smart transportation systems.

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CHAPTER – 1

1.INTRODUCTION

1.1 PROBLEM STATEMENT

Railway transportation plays a vital role in modern infrastructure, carrying millions of passengers and goods every day. However, railway tunnels, especially those located in remote or hilly regions, face several operational and safety challenges. One of the major issues is the **availability of reliable power supply** for tunnel lighting systems. Conventional tunnel lighting relies on continuous electrical power from the grid, which leads to high installation costs, energy consumption, and maintenance requirements. In many remote areas, providing uninterrupted power supply becomes difficult due to geographical constraints and infrastructure limitations.

Another critical issue is **poor visibility inside tunnels**, which directly affects the safety of loco pilots and train operations. Insufficient lighting can lead to reduced reaction time, difficulty in identifying obstacles, and increased risk during emergency situations. Although continuous lighting can solve visibility issues, it results in unnecessary energy consumption even when no train is present in the tunnel.

At the same time, a significant amount of **mechanical and aerodynamic energy is generated and wasted** during train movement. When a train travels through a tunnel, it produces strong vibrations due to the interaction between wheels and tracks, as well as structural oscillations. Additionally, as the train enters and exits the tunnel, it creates high-pressure airflow due to sudden changes in air displacement. These energy sources are naturally available but are not utilized effectively in existing railway systems.

Existing solutions such as grid-powered lighting and solar energy systems have their own limitations. Grid-based systems depend entirely on external power sources and involve high operational costs, while solar energy is not effective inside tunnels due to lack of sunlight. Moreover, current systems do not focus on utilizing the **wasted energy generated by the train itself**, which represents a potential renewable energy source.

Therefore, there is a clear need for an innovative system that can **harvest energy from train-induced vibration and airflow**, convert it into usable electrical energy, and utilize it for practical applications such as tunnel lighting. The system should be self-sustaining, cost-effective, and capable of operating automatically without external power dependency. By addressing these challenges, the proposed solution aims to improve energy efficiency, reduce operational costs, and enhance safety in railway tunnel environments.

1.2 CUSTOMERS & STAKE HOLDERS

Railway Authorities

The major customer for this system is Indian Railways and similar railway bodies. These authorities are responsible for maintaining tunnels, tracks, and safety systems. The proposed system helps them reduce electricity consumption, lower operational costs, and improve tunnel safety by providing a self-powered lighting solution.

Metro Rail Corporations

Urban metro systems such as Delhi Metro Rail Corporation are important customers. Metro trains operate frequently in underground tunnels, making them suitable for continuous energy harvesting. This system can help metro authorities reduce dependency on external power and promote sustainable urban transportation.

Infrastructure Development Companies

Companies involved in railway construction and tunnel development can adopt this system during the design phase. By integrating energy harvesting technology into infrastructure, they can reduce the need for separate power installations and create more efficient and eco-friendly railway systems.

Government and Renewable Energy Bodies

Government organizations and energy management agencies focusing on renewable energy can support and implement this system as part of green initiatives. The project aligns with sustainable development goals by utilizing wasted energy and reducing environmental impact.

1.3 OBJECTIVES

- **To Harvest Energy from Train-Induced Vibration**
 - Train movement creates vibrations on railway tracks. The objective is to capture this vibration using piezoelectric sensors and convert it into electrical energy.
- **To Generate Energy from Airflow in Tunnels**
 - When a train moves through a tunnel, it produces strong airflow. This airflow is used to rotate micro wind turbines to generate additional electrical energy.
- **To Develop a Hybrid Energy System**
 - The main objective is to combine both vibration and airflow energy sources into a single hybrid system to increase overall power generation and reliability.
- **To Store and Manage Generated Energy Efficiently**
 - The generated energy is stored in rechargeable batteries using proper charging circuits, ensuring continuous power availability even when trains are not passing.
- **To Provide Self-Powered Tunnel Lighting System**
 - The final objective is to use the stored energy to automatically power LED lights in tunnels, improving visibility and reducing dependence on external power supply.

CHAPTER – 2

2.SURVEY STUDIES

2.1 CUSTOMER SURVEY

- Most railway staff from organizations like Indian Railways reported lack of reliable power supply in tunnel regions, especially in remote areas.
- Respondents highlighted that conventional lighting systems consume high energy and increase operational and maintenance costs.
- Poor visibility inside tunnels was identified as a major safety concern affecting loco pilots and maintenance workers.
- Continuous lighting systems were found to be inefficient as they waste energy when no train is present.
- Survey participants noted that vibration and airflow energy generated by trains are currently unused, indicating a potential energy source.
- Most stakeholders showed interest in a self-sustaining, automatic energy harvesting system that can reduce dependency on external power and improve safety.

2.2 LITERATURE SURVEY

- A review of vibration energy harvesting in rail transportation field
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Author: Peyman Aela, Hung-Lin Chi*, Yiwei Weng, Yangsheng Lin and David P Connolly
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Author: Muhammad Iqbal
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- Piezoelectric energy harvesting in railways: Current status and future challenges
Author: Ding, Dong Fan, Jiahe ;Wang, Haoyu ;Jing, Guoqing ;Guo, Yunlong
Link: <https://ui.adsabs.harvard.edu/abs/2025EneSR..6101869D/abstract#:~:text=Piezoelectric%20energy%20harvesting%2C%20which%20converts,design%2C%20a nd%20ease%20of%20integration.>

2.3 PATENT SURVEY

- **Method and system for energy harvesting**
Inventor: Richard James,Walter Mansfield,Daniel Shani,Nissim Shani
Link: <https://patents.google.com/patent/US8344529B2/en>
- **Energy harvesting system, apparatus and method**
Inventor: Mahadevan Balasubramaniam,Jeffrey Bernard Fortin,Walter John Smith,Huageng Luo
Link: <https://patents.google.com/patent/US7116036B2/en>
- **Vibration energy harvesting and management for wireless sensor networks in bridge structural monitoring**
Inventor: Wei Liu, Zhengqiang Wang, Rong Luo, Shaohua Qu
Link: <https://ieeexplore.ieee.org/document/7370561/authors#authors>
- **Energy Harvesting From Loudspeaker Vibration**
Inventor: TELCI,Emre
Link: https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2020055367&_cid=P22-MMPVAF-04397-1

2.4 MARKET SURVEY

- Existing railway tunnel lighting systems are mainly grid-powered, leading to high installation, operation, and maintenance costs.
- Renewable solutions like solar energy have limited use in tunnels due to lack of sunlight and space constraints.
- Current market solutions do not focus on utilizing train-generated vibration and airflow energy, leaving a gap for innovation.
- There is increasing demand for energy-efficient, low-cost, and self-sustaining systems in modern railway infrastructure.

2.5 EXISTING PROBLEMS

- **Conventional Tunnel Lighting Systems**
Uses electricity from the power grid with fixed lights installed inside the tunnel.
Consumes more energy and requires high wiring and maintenance cost.
- **Basic Energy Harvesting Systems**
Some systems use single-source harvesting (only vibration or only airflow), but not hybrid integration.
- **Battery Energy Storage Systems (BESS)**
Large battery banks store external electricity and supply power to tunnel lights when required.
- **Smart LED Lighting Systems**
Uses energy-efficient LED lights with automatic control systems. Reduces power consumption but still depends on external electricity supply.

CHAPTER – 3

3.PROPOSED WORK

3.1 SYSTEM ARCHITECTURE

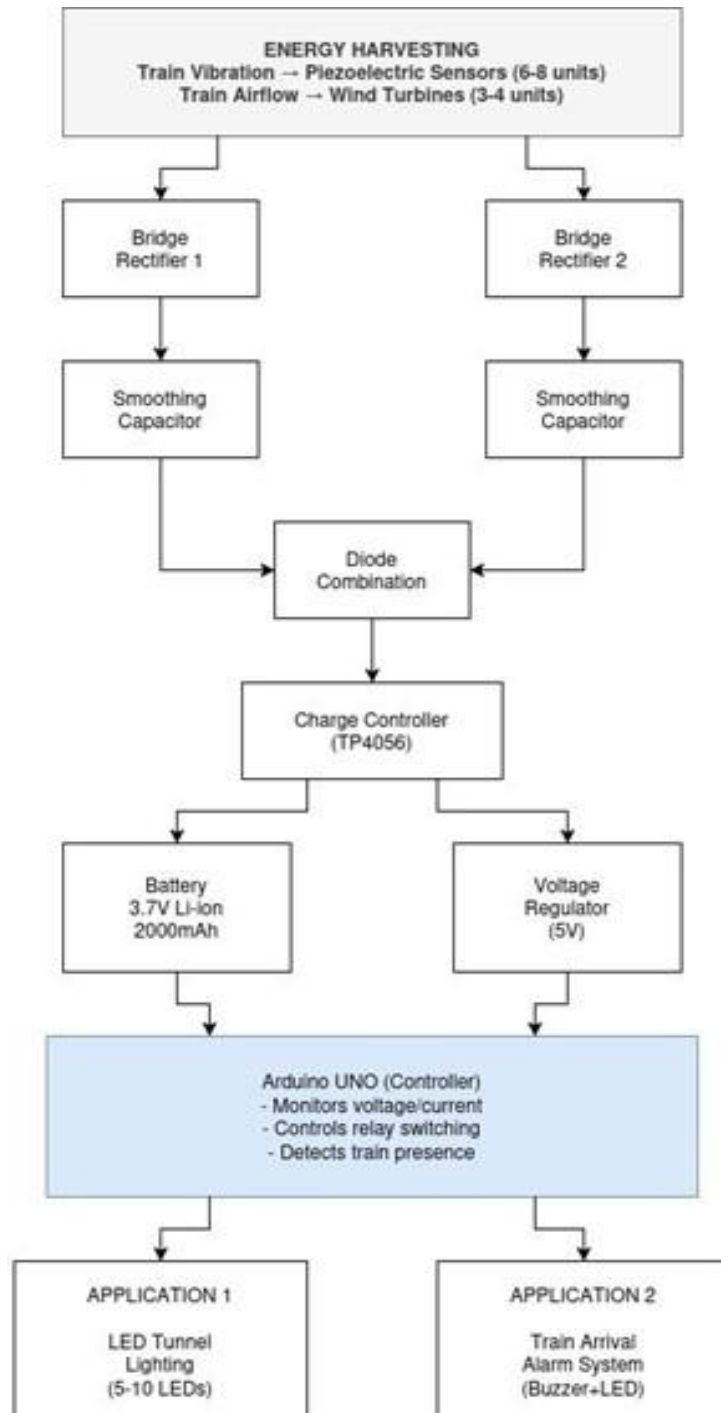


Fig: 3.1 System Architecture

3.1.1 Block Diagram Explanation

- **Energy Sources**
 - Piezo sensors (6–8 units) → generate voltage from vibration
 - Wind turbines (3–4 units) → generate AC from airflow
- **Bridge Rectifier**
 - Converts AC → DC
 - Uses 4 diodes (1N4007 in your circuit)
- **Smoothing Capacitor**
 - Removes fluctuations (ripples)
 - Stores small charge temporarily
- **Diode Combination (Critical Part)**
 - Combines outputs from piezo + turbine
 - Prevents reverse current flow
- **TP4056 Charge Controller**
 - Charges 3.7V Li-ion battery
 - Protects from:
 - Overcharge
 - Over-discharge
- **Battery (3.7V, 2000mAh)**
 - Stores harvested energy
 - Acts as backup when no train is present
- **Voltage Regulator (5V)**
 - Boosts battery voltage → stable 5V
 - Needed for Arduino UNO

- **Arduino UNO**

Main brain:

- Monitors voltage
- Detects train (via ultrasonic)
- Controls relay / outputs

- **Applications**

- LED tunnel lighting
- Train arrival alert (buzzer + LED)

3.2 FLOW CHART

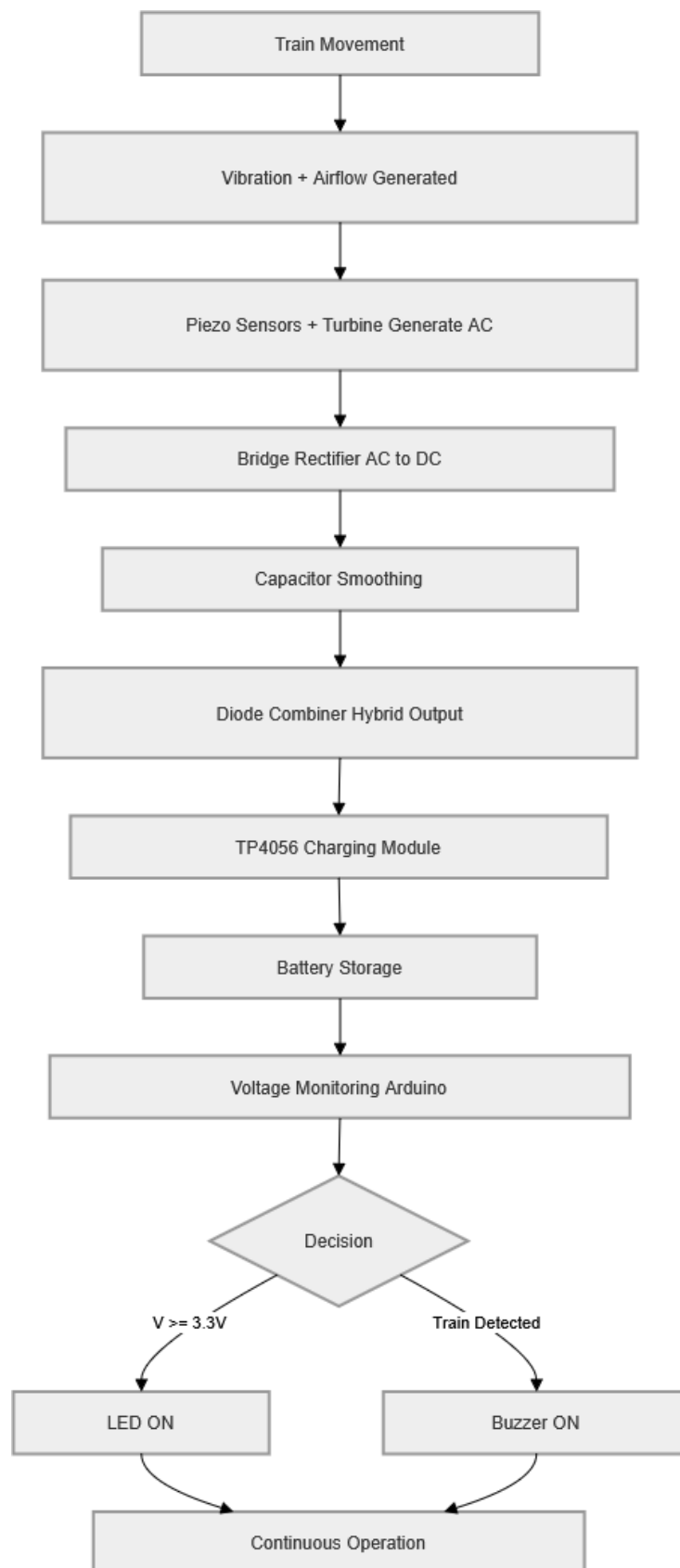


Fig: 3.2 Flow chart

3.2.1 Flow logic:

1. Train moves
2. Generates vibration + airflow
3. Piezo + turbine → AC
4. Rectifier → DC
5. Capacitor → smooth DC
6. Diode → combine sources
7. TP4056 → charge battery
8. Arduino monitors voltage

3.3 EXPERIMENTAL SETUP

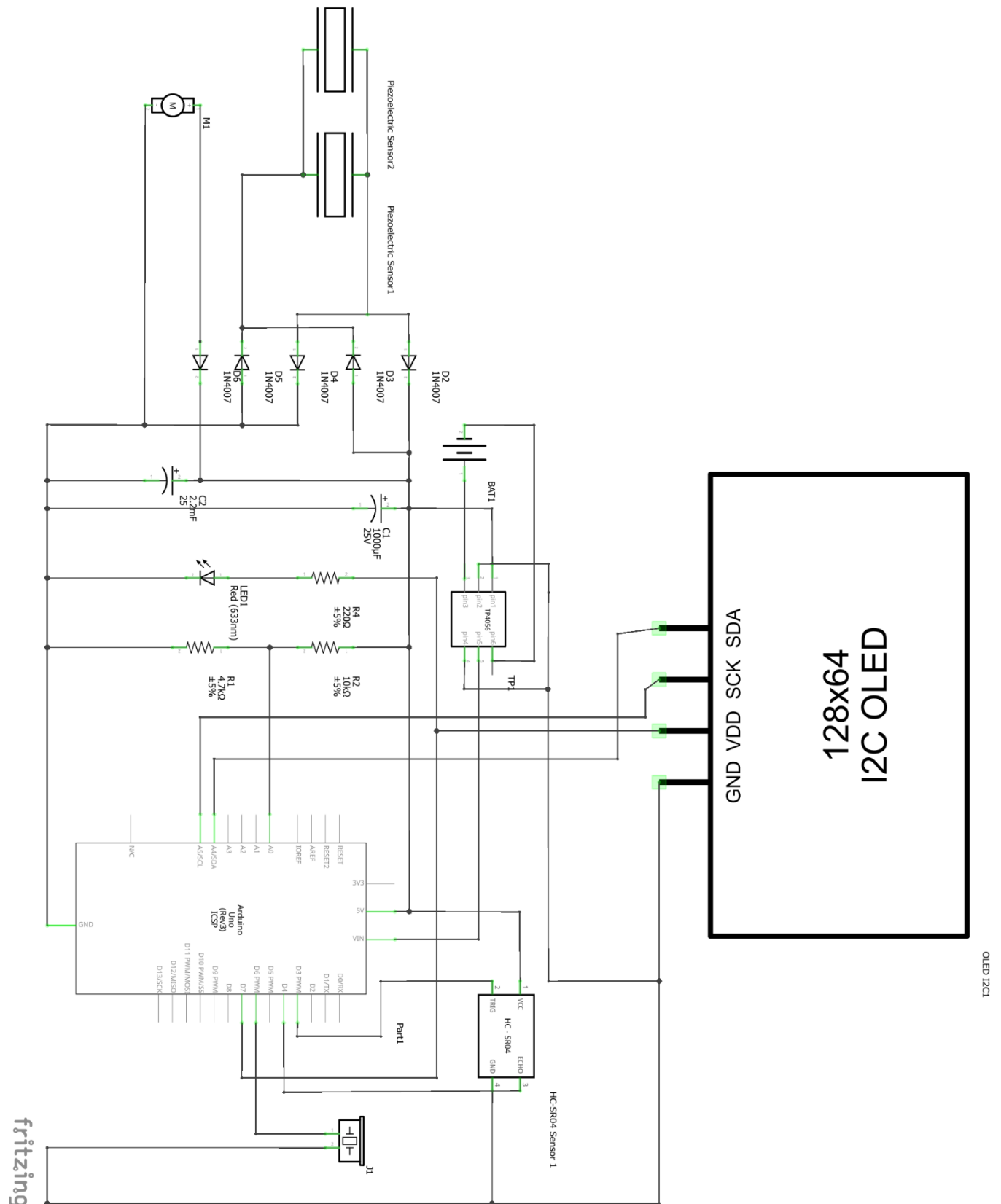


Fig: 3.3 Experiment Setup

3.3.1 Circuit Diagram Explanation

1. Piezoelectric Sensor Section

Multiple piezoelectric sensors are used as the primary energy source. These sensors generate AC voltage when subjected to mechanical vibrations caused by train movement.

- The sensors are connected in a suitable combination (series/parallel) to increase voltage/current output.
- Since the output is irregular AC, it cannot be directly used.

2. Bridge Rectifier (Diodes: 1N4007)

The AC output from the piezo sensors is fed into a bridge rectifier made using four diodes (1N4007).

Function:

- Converts AC $\rightarrow$ pulsating DC
- Ensures current flows in only one direction

This stage is essential because all further components require DC supply.

3. Smoothing Capacitors

Capacitors (e.g., 1000 μ F, 25V) are connected after the rectifier.

Function:

- Reduce voltage ripple
- Store charge temporarily
- Provide a more stable DC output

Without capacitors, the voltage would fluctuate heavily.

4. Diode Combination Network

Outputs from multiple sources (piezo + turbine if included) are combined using diodes.

Function:

- Prevent reverse current flow between sources
- Allow safe merging of multiple energy inputs

This ensures one source does not discharge into another.

5. Battery and Charging Module (TP4056)

The combined DC output is fed into a TP4056 charging module, which charges a 3.7V Li-ion battery.

Function:

- Controls charging voltage and current
- Provides protection:
 - Overcharging
 - Over-discharging
 - Short circuit

The battery acts as an energy storage unit to supply power when generation is low.

6. Voltage Regulation

A voltage regulator (boost converter or 5V regulator) is used to step up or stabilize the battery voltage.

Function:

- Converts battery voltage (~3.7V) to stable 5V
- Required for Arduino UNO operation

7. Arduino UNO (Controller)

The Arduino UNO is the central control unit.

Functions:

- Monitors voltage level using analog input
- Controls outputs such as LEDs and buzzer

- Processes signals from sensors

8. OLED Display (128×64, I2C)

The OLED display is connected using I2C communication.

Connections:

- VCC → 5V
- GND → GND
- SDA → A4
- SCL → A5

Function:

- Displays voltage readings
- Shows system status

9. Ultrasonic Sensor (HC-SR04)

Used for detecting train presence.

Connections:

- VCC → 5V
- GND → GND
- Trigger → Digital pin
- Echo → Digital pin

Function:

- Measures distance
- Detects approaching train

10. Output Section (LED and Buzzer)

LED:

- Used for lighting (e.g., tunnel lighting)

- Controlled via Arduino

Buzzer:

- Activated when a train is detected
- Provides an alert signal

11. Working Summary

1. Piezo sensors generate AC voltage from vibration
2. Bridge rectifier converts it to DC
3. Capacitor smooths the DC output
4. Diodes combine multiple energy sources
5. TP4056 charges the Li-ion battery
6. Voltage regulator provides stable 5V
7. Arduino monitors and controls the system
8. Outputs (LED, buzzer) operate based on conditions

CHAPTER – 4

4.RESULTS AND DISCUSSION

4.1 Experimental Results

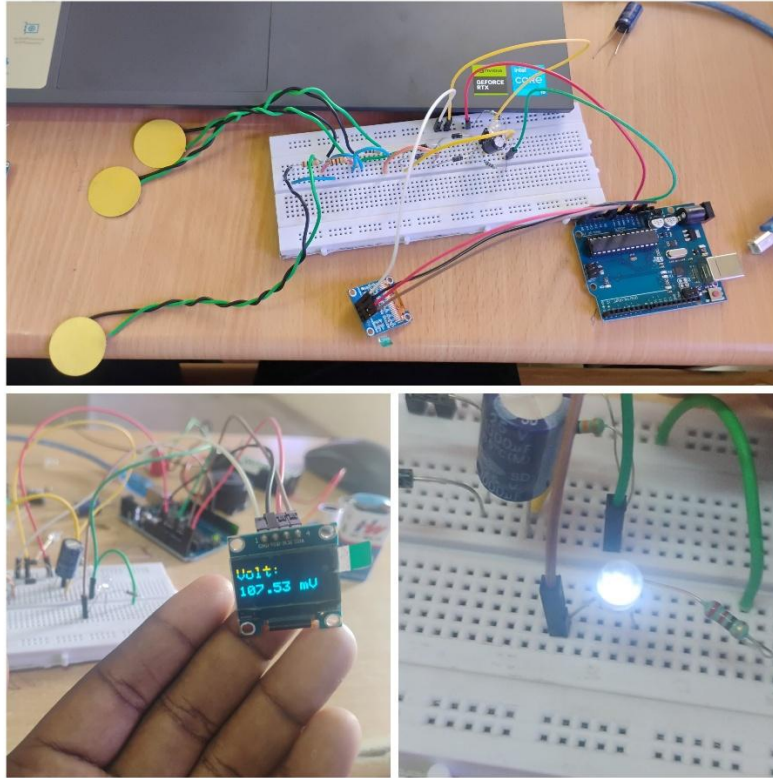


Fig 4.1: Experimental Results

The performance of the hybrid system was evaluated through a series of tests targeting individual energy sources and the integrated system.

- **Piezoelectric Energy Generation Test:**
 - **Objective:** To measure the voltage generated from simulated train-induced vibrations.
 - **Outcome:** The piezoelectric sensors successfully converted mechanical pressure into electrical energy. The raw output voltage ranged from 0.5V to 4V (AC), which, after rectification, provided a stable 0.3V to 3V (DC).
 - **Inference:** While the power output from piezo sensors is relatively low, it remains consistent and suitable for low-power sensor operations.
- **Airflow Energy Generation Test:**

- **Objective:** To evaluate the energy captured by the DC motor-based turbine from airflow.
- **Outcome:** The motor acting as a generator produced a voltage between 1V and 5V (DC) with a current of 10–50 mA.
- **Inference:** The airflow module provides a significantly higher power output compared to the piezoelectric module, confirming its effectiveness in high-airflow tunnel environments.
- **Hybrid Energy Combination & Storage Test:**
 - **Objective:** To verify the efficiency of combining both energy sources and storing them.
 - **Outcome:** The combined output through the diode module reached 2V to 5V (DC). The TP4056 charging module successfully increased the battery voltage from 3.6V to 3.9V during testing.
 - **Inference:** The hybrid approach ensures more stable and continuous energy availability compared to using a single source.

4.2 Discussion

The experimental data confirms that the integration of vibration and airflow harvesting provides a viable self-sustaining power source for railway infrastructure.

System Synergy: The primary advantage of this hybrid system is its ability to utilize two different wasted energy forms. While the piezoelectric sensors capture the immediate mechanical impact of the train on the tracks, the turbine continues to harvest energy from the aerodynamic "piston effect" created as the train moves through the tunnel.

Power Management: The use of bridge rectifiers and smoothing capacitors (1000 μ F – 2200 μ F) was critical in reducing voltage ripple to less than 5%, ensuring the harvested energy is safe for the Arduino microcontroller and charging modules.

Efficiency and Practicality: Total power calculations indicate a combined output of approximately 0.31W under simulated conditions. Given that a standard tunnel LED requires only 0.04W, the system is capable of powering multiple lights simultaneously.

Comparison with Existing Solutions: Unlike solar energy, which is ineffective in tunnels, or grid-based lighting, which suffers from high installation and maintenance costs in remote areas, this system is self-contained and operates automatically whenever a train is present.

Parameter	Piezoelectric Module	Airflow Module	Hybrid System
Voltage (DC)	0.3V – 3V	1V – 5V	2V – 5V
Stability	Low (Vibration dependent)	Medium (Speed dependent)	High (Combined)
Primary Use	Low-power sensing	Battery charging/Lighting	Continuous tunnel power +1

Table: 4.1 Result & Discussion

CHAPTER – 5

5.CONCLUSION & FUTURE SCOPE

5.1 CONCLUSION

The proposed hybrid energy harvesting system demonstrates an effective method of converting train-induced vibration and airflow into usable electrical energy for tunnel lighting applications. The prototype successfully shows that wasted mechanical and aerodynamic energy in railway tunnels can be harnessed using piezoelectric sensors and a turbine-based mechanism. By combining both energy sources, the system improves overall efficiency and reliability compared to single-source approaches. The generated energy is sufficient to power LED lights, enhancing visibility and safety inside tunnels while reducing dependency on external power supply. Overall, the system provides a cost-effective, eco-friendly, and self-sustaining solution, highlighting its potential for real-time implementation in modern railway infrastructure.

5.2 FUTURE DIRECTIONS

- Improve energy generation by using high-efficiency piezoelectric materials and optimized turbine blade design.
- Integrate rechargeable battery storage systems to provide continuous lighting even after the train passes.
- Implement IoT-based monitoring for real-time tracking of energy generation and system performance.
- Optimize sensor placement and tunnel design to maximize vibration and airflow energy capture.
- Scale the system for real-time deployment in railway and metro tunnels to enhance energy efficiency and safety.

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Author: P. Peralta-Braz, [M.M. Alamdari](#) , R.O. Ruiz , [E. Atroshchenko](#)
Link: <https://www.sciencedirect.com/science/article/pii/S0888327023007318>
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Link: <https://www.mdpi.com/2076-0825/10/12/312>